M3403 ELEMENTARY DIFFERENTIAL EQUATIONS

WEEK 1 2020/2021

FIRST ORDER ORDINARY DIFFERENTIAL EQUATIONS

1. SEPARABLE EQUATIONS

A first order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y) \quad \text{——①}$$

is a variable separable equation.

This type of equations are solved by simple integration. Equation ① can be written as

$$\frac{1}{h(y)}\,dy = g(x)\,dx$$

$$\Rightarrow \int \frac{1}{h(y)}\,dy = \int g(x)\,dx$$

$$\Rightarrow H(y) = G(x) + C$$

Examples

1. Solve $e^x y' + x(1+y^2) = 0$, $y(-1) = 0$
2. If $P(t)$ is the amount of dollars in a savings bank account that pays yearly interest rate of $r\%$ compounded continuously, then

$$\frac{dP}{dt} = \frac{r}{100}P \; ; \quad t \text{ in years}$$

Assume that the interest rate is 5% annually, $P(0) = 1000$ and no withdrawals are made, how much will be in the account after 2 years?

3. Driver A had been leading driver B for a while by a steady 3 miles. Only 2 miles from the finish line, driver A ran out of gas and decelerated thereafter at a rate proportional to the square of his remaining speed. One mile later, driver A's speed was exactly halved. If driver B's speed remained constant, who won the race?

Sol'n

① $e^x y' + x(1+y^2) = 0$, $y(-1) = 0$

$\Rightarrow \frac{1}{1+y^2}\, dy = -xe^{-x}\, dx$

$\Rightarrow \tan^{-1} y = xe^{-x} + e^{-x} + C$

$\tan^{-1}(0) = -e + e + C$

$\Rightarrow \quad 0 = C$

$\Rightarrow \tan^{-1} y = xe^{-x} + e^{-x}$ //

② $\frac{dP}{dt} = \frac{r}{100} P$

$\frac{1}{P} dP = \frac{r}{100} dt$

$\Rightarrow \ln P = \frac{tr}{100} + C$

$\ln 1000 = C$

$\therefore \ln P = \frac{tr}{100} + \ln 1000$

$\Rightarrow \ln \frac{P}{1000} = \frac{tr}{100}$

$\Rightarrow \quad P = 1000 \cdot e^{\frac{tr}{100}}$

$$P(2) = 1000\, e^{\frac{2\times 5}{100}}$$

$$\simeq 1105.1709$$

∴ $1105.17 will be in the account

(5) B ——— A ———
3 miles ⋮ 2 miles
t=0

$v(0) = v_0$ for both cars since they were travelling at same speed before

For B

$x(0) = -3$

Since B is travelling at constant speed

$$\frac{dv}{dt} = 0$$

$$\Rightarrow v(t) = C_1$$

$$v(0) = v_0 = C_1$$

$$\therefore v(t) = v_0$$

$$\Rightarrow \frac{dx}{dt} = v_0$$

$$\Rightarrow x = v_0 t + C_2$$

$$x(0) = C_2 = -3$$

$$\therefore x(t) = v_0 t - 3$$

Time taken to cover 2 miles / finish

$$2 = v_0 t - 3$$

$$\Rightarrow t = \frac{5}{v_0}$$

∴ After A ran out of gas, it took B $t = \frac{5}{v_0}$ to finish the race

For A

$x(0) = 0$

$$\frac{dv}{dt} = -kv^2$$

$$\Rightarrow -\frac{dv}{v^2} = k\,dt$$

$$\Rightarrow \frac{1}{v} = kt + C_3$$

$$\frac{1}{v_0} = k(0) + C_3$$

$$\Rightarrow \frac{1}{v_0} = C_3$$

so $\frac{1}{v} = kt + \frac{1}{v_0}$

$$\Rightarrow v = \frac{v_0}{kt v_0 + 1} \quad -(1)$$

$$\frac{dx}{dt} = \frac{v_0}{kt v_0 + 1}$$

$$\Rightarrow x(t) = \frac{1}{k}\ln(kt v_0 + 1) + C_4$$

$$x(0) = \frac{1}{k}\ln 1 + C_4 = 0$$

$$\Rightarrow C_4 = 0$$

$$\therefore x(t) = \frac{1}{k}\ln(kt v_0 + 1) \quad -(2)$$

After 1 mile, time taken was

$$1 = \frac{1}{k}\ln(kt v_0 + 1)$$

$$\Rightarrow e^k = kt v_0 + 1$$

$$\Rightarrow t = \frac{e^k - 1}{k v_0}$$

At this time $v(t) = \frac{1}{2}v_0$, so

$$\frac{1}{2}v_0 = \frac{v_0}{v_0 k\left(\frac{e^k - 1}{k v_0}\right) + 1}$$

$$\Rightarrow e^k = 2$$

$$\Rightarrow k = \ln 2$$

Substituting in (2)

$$x(t) = \frac{1}{\ln 2}\ln(t v_0 \ln 2 + 1)$$

Time taken to finish the remaining 2 miles is

$$2 = \frac{1}{\ln 2}\ln(t v_0 \ln 2 + 1)$$

$$\Rightarrow 4 = t v_0 \ln 2 + 1$$

$$\Rightarrow t = \frac{3}{\ln 2\, v_0}$$

$$\simeq \frac{4.328}{v_0}$$

since $\frac{4.328}{v_0} < \frac{5}{v_0}$

A wins the race

2. Linear Equations

These equations can be expressed in the form

$$\frac{dy}{dx} + P(x)y = f(x) \quad \text{— (1)}$$

To solve the linear equation, we

- Express the linear equation in standard form (eqn (1))
- From the standard form, identify $P(x)$, the find the integrating factor $e^{\int P(x)dx}$, the constant of integration is taken as zero.
- Multiply both sides by the integrating and then integrate

Examples

(1) Solve $x\frac{dy}{dx} + 3(y+x^2) = \frac{\sin x}{x}$

(2). A rock contains two radioactive isotopes RA_1 and RA_2, that belong to the same radioactive series, that is; RA_1 decays into RA_2, which then decays into stable atoms. Assume that the rate at which RA_1 decays into RA_2 is $50e^{-10t}$ kg/sec, and the rate of decay of RA_2 is proportional to the mass $y(t)$ of RA_2 present, the rate of change in RA_2 is

$\frac{dy}{dt}$ = rate of creation — rate of decay

i.e $\frac{dy}{dt} = 50e^{-10t} - ky$

Where $k>0$ is the decay constant.

Find the mass of RA_2 for $t>0$, if $k = 2$/sec and the initial mass of RA_2 is 40kg

Soln

① $x\frac{dy}{dx} + 3(y+x^2) = \frac{\sin x}{x}$

$\Rightarrow \frac{dy}{dx} + \frac{3y}{x} = \frac{\sin x}{x^2} - 3x$

Integrating factor $= \exp\left(\int \frac{3}{x}\,dx\right)$

$= \exp(3\ln x)$

$= x^3$

$x^3\frac{dy}{dx} + 3x^2y = x\sin x - 3x^4$

$\frac{d}{dx}(x^3y) = x\sin x - 3x^4$

$x^3y = -x\cos x + \sin x - \frac{3}{5}x^4 + C$

② $\frac{dy}{dt} = 50e^{-10t} - Ky$

$\Rightarrow \frac{dy}{dt} + Ky = 50e^{-10t}$

Integrating factor $= e^{\int K\,dt}$

$= e^{Kt}$

$e^{Kt}\frac{dy}{dt} + Ke^{Kt}y = 50e^{Kt-10t}$

$\frac{d(e^{Kt}y)}{dt} = 50e^{t(K-10)}$

$e^{Kt}y = \frac{50}{K-10}e^{t(K-10)} + C, \quad K \neq 10$

$e^{2(0)}\cdot 40 = \frac{50}{2-10}e^{0} + C$

$C = 40 + \frac{50}{8} = \frac{370}{8} = \frac{185}{4}$

$y = \frac{-50}{8}e^{-8t-2t} + \frac{185}{4}e^{-2t} = \frac{185}{4}e^{-2t} - \frac{25}{4}e^{-10t}, \quad t \geq 0$

3 Exact Equations

The differential equation $M(x,y)dx + N(x,y)dy = 0$ is exact if there exist a function $F(x,y)$ such that $f_x = M(x,y)$ and $f_y = N(x,y)$

Test For Exactness

The equation $M(x,y)dx + N(x,y)dy = 0$ is exact if and only if the condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \forall (x,y)$$

holds

Method For solving Exact Equations

1. Test for exactness, ie check if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
2. If the equation is exact, then there exists F such that

$$\frac{\partial F}{\partial x} = M(x,y)$$

$$\Rightarrow F = \int M(x,y)dx + h(y) \quad \text{---} \quad *$$

3. To find $h(y)$, differentiate F with respect to y and equate it to $N(x,y)$ since $F_y = N$

4. Solve for $h'(y)$ and integrate to get $h(y)$, substistute it into $*$

The general soln is given by

$$F(x,y) = C$$

Examples

① Solve $(2xy+3)dx + (x^2-1)dy = 0$

Soln

Let $M = 2xy+3$ and $N = x^2-1$

$M_y = 2x$ $\qquad$ $N_x = 2x$

$M_y = N_x$

let $F_x = 2xy+3$

$F = x^2y + 3x + g(y)$

$F_y = x^2 + g'(y) = x^2 - 1$

$\Rightarrow g'(y) = -1$

$\Rightarrow g(y) = -y + k$

$\therefore x^2y + 3x - y = C$ is the soln

A Non-Exact Equation Made Exact

If $M(x,y)dx + N(x,y)dy = 0$ is not exact, but $\mu(x,y)M(x,y)dx + \mu(x,y)N(x,y)dy = 0$ is exact, then μ is called the integrating factor of $Mdx + Ndy = 0$.

Since the second equation is exact, we have

$$\frac{\partial}{\partial y}\left(\mu(x,y)M(x,y)\right) = \frac{\partial}{\partial x}\left(\mu(x,y)N(x,y)\right)$$

$$\Rightarrow \mu\frac{\partial M}{\partial y} + M\frac{\partial \mu}{\partial y} = \mu\frac{\partial N}{\partial x} + N\frac{\partial \mu}{\partial x}$$

$$\Rightarrow M\frac{\partial \mu}{\partial y} - N\frac{\partial \mu}{\partial x} = \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right]\mu \quad \text{—— (1)}$$

Solving eq^{n} ① for u can be difficult except if $u = u(x)$ or $u = u(y)$

Lets assume that $u = u(x)$, then from ①, we get

$$\frac{du}{dx} = \left(\frac{My - Nx}{N}\right) u \quad \text{where } My - Nx = f(x)$$

Similarly if $u = u(x)$, then

$$\frac{du}{dy} = \left(\frac{Nx - My}{M}\right) u \quad \text{where } Nx - My = f(y)$$

If we take $u = u(x)$, we will have

$$\frac{1}{u} du = \left(\frac{My - Nx}{N}\right) dx$$

$$\Rightarrow \ln u = \int \left(\frac{My - Nx}{N}\right) dx$$

$$\Rightarrow u = \exp\left(\int \frac{My - Nx}{N} dx\right)$$

Similarly if $u = u(y)$, then

$$u = \exp\left(\int \frac{Nx - My}{M} dy\right)$$

Examples

Solve ① $xy\,dx + (2x^2 + 3y^2 - 20)\,dy = 0$

② $(2x^2 + y)\,dx + (x^2 y - x)\,dy = 0$

Sol^{n}:-

① . $xy\,dx + (2x^2+3y^2-20)dy = 0$

let $M = xy$ and $N = 2x^2+3y^2-20$

$M_y = x$, $N_x = 4x$

$M_y \neq N_x$, thus the eqⁿ is not exact.

$\frac{M_y - N_x}{N} = \frac{x-4x}{2x^2+3y^2-20} = \frac{-3x}{2x^2+3y^2-20} \neq f(x)$, we can't use it.

$\frac{N_x - M_y}{m} = \frac{4x-x}{xy} = \frac{3}{y} = f(y)$, so we can use this to find the integrating factor

$u = \exp \int \frac{3}{y}\,dy$

$= \exp(3\ln y)$

$= y^3$

Multiplying by u we get

$xy^4\,dx + (2x^2y^3+3y^5-20y^3)dy = 0$

$M_y = 4xy^3$ $N_x = 4xy^3$

$M_y = N_x$

Let $F_x = xy^4$

$\Rightarrow F = \frac{1}{2}x^2y^4 + g(y)$

$\Rightarrow F_y = 2x^2y^3 + g'(y) = 2x^2y^3 + 3y^5 - 20y^3$

$\Rightarrow g'(y) = 3y^5 - 20y^3$

$\Rightarrow g(y) = \frac{1}{2}y^6 - 5y^4 + K$

The soln is $\frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = C$ //

② $(2x^2+y)\,dx + (x^2y - x)\,dy = 0$

let $M = 2x^2+y$, $N = x^2y - x$

$M_y = 1$ $\qquad$ $N_x = 2xy - 1$

$M_y \neq N_x$

$$\frac{M_y - N_x}{N} = \frac{1 - 2xy + 1}{x^2y - x} = \frac{2(1-xy)}{x(xy-1)} = -\frac{2}{x} = f(x)$$

$$u = \exp\left(\int -\frac{2}{x}\,dx\right)$$
$$= \exp(-2\ln x)$$
$$= x^{-2}$$

Multiplying by u, we get

$(2 + yx^{-2})\,dx + (y - x^{-1})\,dy = 0$ which is exact

let $F_x = 2 + yx^{-2}$

$\Rightarrow F = 2x - yx^{-1} + g(y)$

$\Rightarrow F_y = -x^{-1} + g'(y) = y - x^{-1}$

$\Rightarrow g'(y) = y$

$\Rightarrow g(y) = \frac{1}{2}y^2 + K$

$\therefore 2x - yx^{-1} + \frac{1}{2}y^2 = C$ is the soln

4. Homogeneous Equations

If $f(kx, ky) = k^n f(x,y)$, then $f(x,y)$ is a homogeneous funtion of degree n.

A differential equation

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

is a homogeneous equation if M and N are homogeneous functions of the same degree.

To solve a homogeneous equation, we make a substitution $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$.

This substitution transforms a homogeneous equation into a variable separable equation

Example

Solve $(xy + y^2 + x^2)\,dx - x^2\,dy = 0$

Solⁿ $\frac{dy}{dx} = \frac{xy + y^2 + x^2}{x^2}$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{x^2v + x^2v^2 + x^2}{x^2}$$

$$\Rightarrow v + x\frac{dv}{dx} = v + v^2 + 1$$

$$\Rightarrow x\frac{dv}{dx} = v^2 + 1$$

$$\Rightarrow \frac{1}{v^2+1} = \frac{1}{x}dx$$

$$\Rightarrow \tan^{-1}v = \ln x + C$$

$$\Rightarrow y = \tan(\ln x + C)$$

5. Differential Equations Of The Form

$$\frac{dy}{dx} = F(ax+by)$$

This type of equation is solved by making the substitution

$$z = ax+by$$

which transforms the equation into a variable separable equation

Examples

Solve ① $\frac{dy}{dx} = \sqrt{x+y} - 1$

② $\frac{dy}{dx} = y - x - 1 + (x-y+2)^{-1}$

Soln

① Let $z = x+y$

$$\Rightarrow \frac{dz}{dx} = 1 + \frac{dy}{dx} \quad \Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 1$$

$$\frac{dz}{dx} - 1 = \sqrt{z} - 1 \quad , z \geq 0 \quad \text{if } z = 0 \Rightarrow x+y = 0$$

$$\Rightarrow \frac{dz}{\sqrt{z}} = dx \qquad \Rightarrow y = -x$$

$$\Rightarrow 2\sqrt{z} = x + C$$

$$\Rightarrow 2\sqrt{x+y} = x + C$$

$$\Rightarrow \therefore y = \left(\frac{x+C}{2}\right)^2 - x \quad \text{or} \quad y = -x$$

② $\frac{dy}{dx} = y - x - 1 + (x - y + 2)^{-1}$

$= y - x - 1 + (-(y-x) + 2)^{-1}$

let $z = y - x$

$\Rightarrow \frac{dz}{dx} = \frac{dy}{dx} - 1$

$\frac{dz}{dx} + 1 = z - 1 + (2 - z)^{-1}$

$\Rightarrow \frac{dz}{dx} = z - 2 - \frac{1}{z-2}$

$= \frac{(z-2)^2 - 1}{z-2}$

$\Rightarrow \frac{z-2}{(z-2)^2 - 1} dz = dx$

$\Rightarrow \frac{1}{2} \ln\left|(z-2)^2 - 1\right| = x + C$

$\Rightarrow (z-2)^2 - 1 = e^{2x+2C}$

$\Rightarrow (y - x - 2)^2 = Ce^{2x} + 1$ or $(x - y + 2)^2 = Ce^{2x} + 1$

let $u = (z-2)^2$

$\Rightarrow du = 2(z-2)dz$

$\Rightarrow \frac{1}{2} du = (z-2)dz$

$\frac{1}{2}\int \frac{du}{u-1} = \frac{1}{2}\ln|u-1|$

which will be in the account after a year.